MetaboPrep Data Preparation Summary Report

27 March, 2025

Contents

1 Project Information

This `metaboprep2` report summarizes the data preparation steps for:

- **Project:** MYPROJECT
- **Platform:** metabolon_v1
- **Data processing date:** 2025-03-26

1.1 Overview

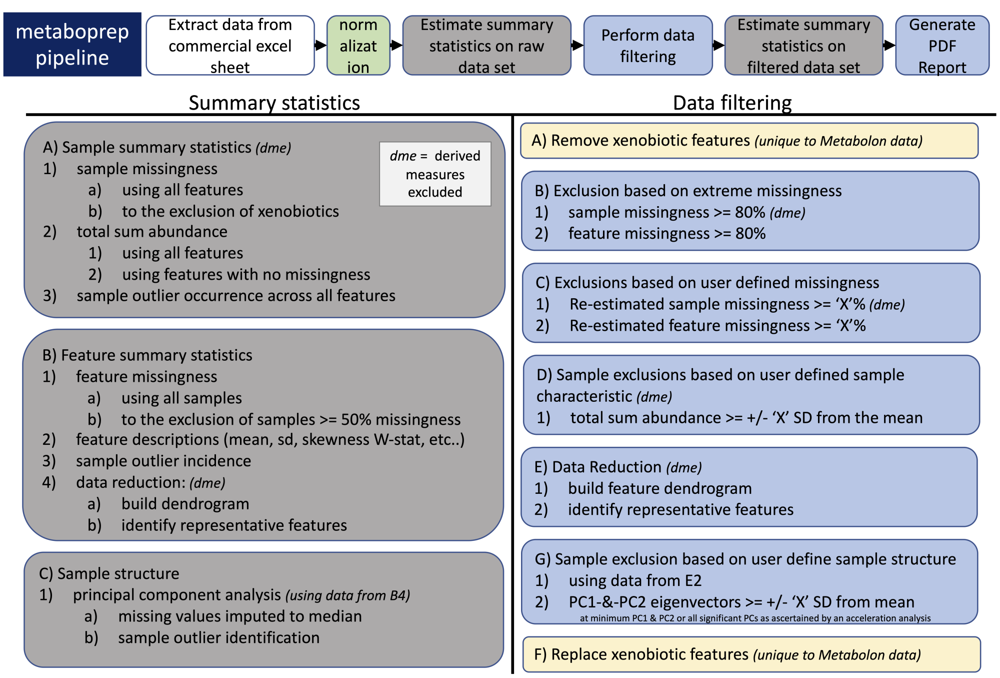
The `metaboprep` R package performs three key operations:

1. **Assessment & Summary Statistics**: Provides an initial assessment of raw metabolomics data.
2. **Data Filtering**: Applies filtering techniques to clean the dataset.
3. **Post-Filtering Assessment**: Evaluates the filtered dataset, particularly in relation to batch variables when available.

This report contains descriptive information on both raw and filtered metabolomics data for **MYPROJECT**.

Please raise any issues on the GitHub issues page.

1.2 Data preparation workflow:



2 Summary of raw data

2.1 Sample size of MYPROJECT data set:

Table 1: Sample Size Table

Dataset	Samples	Features
RAW	100	100
SCALED	100	100
POST QC	98	97

3 Missingness

Missingness is evaluated across samples and features using the original/raw data set.

3.1 Visual structure of missingness in the raw data

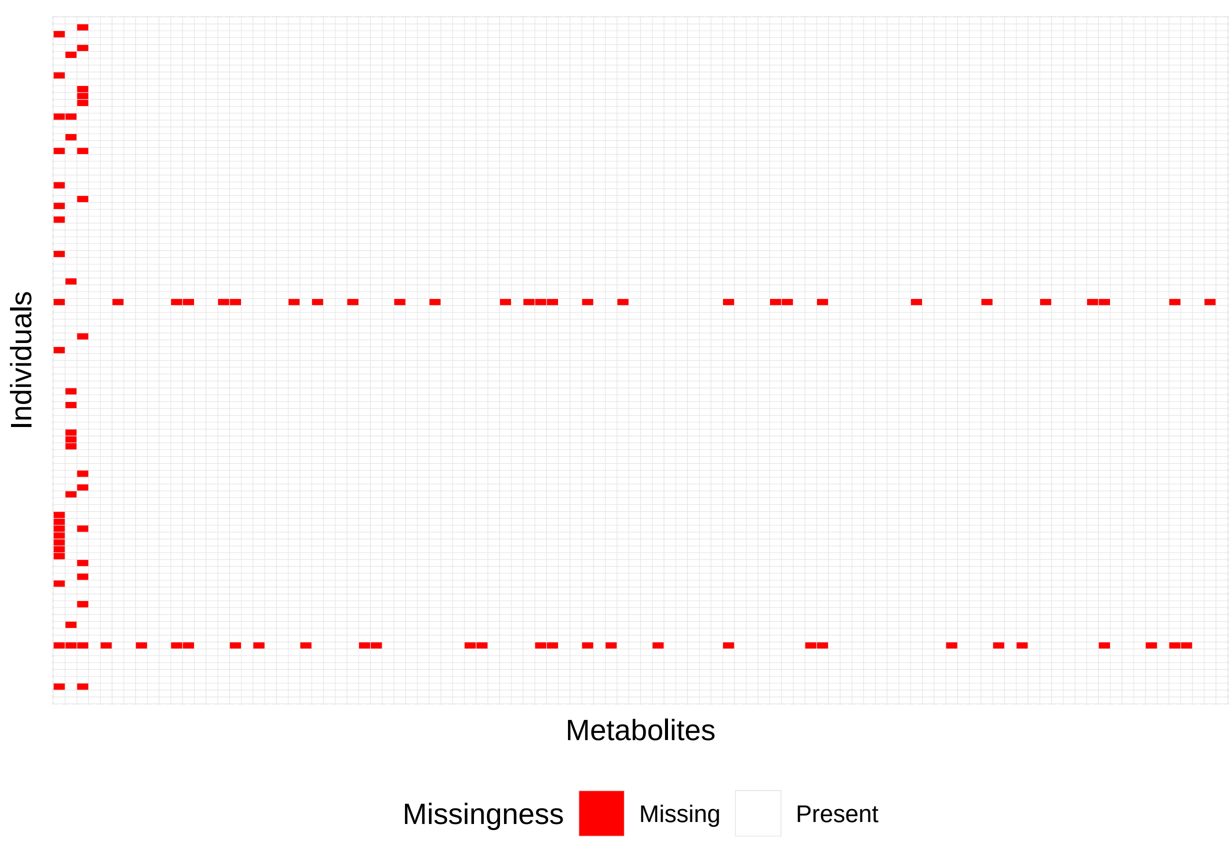


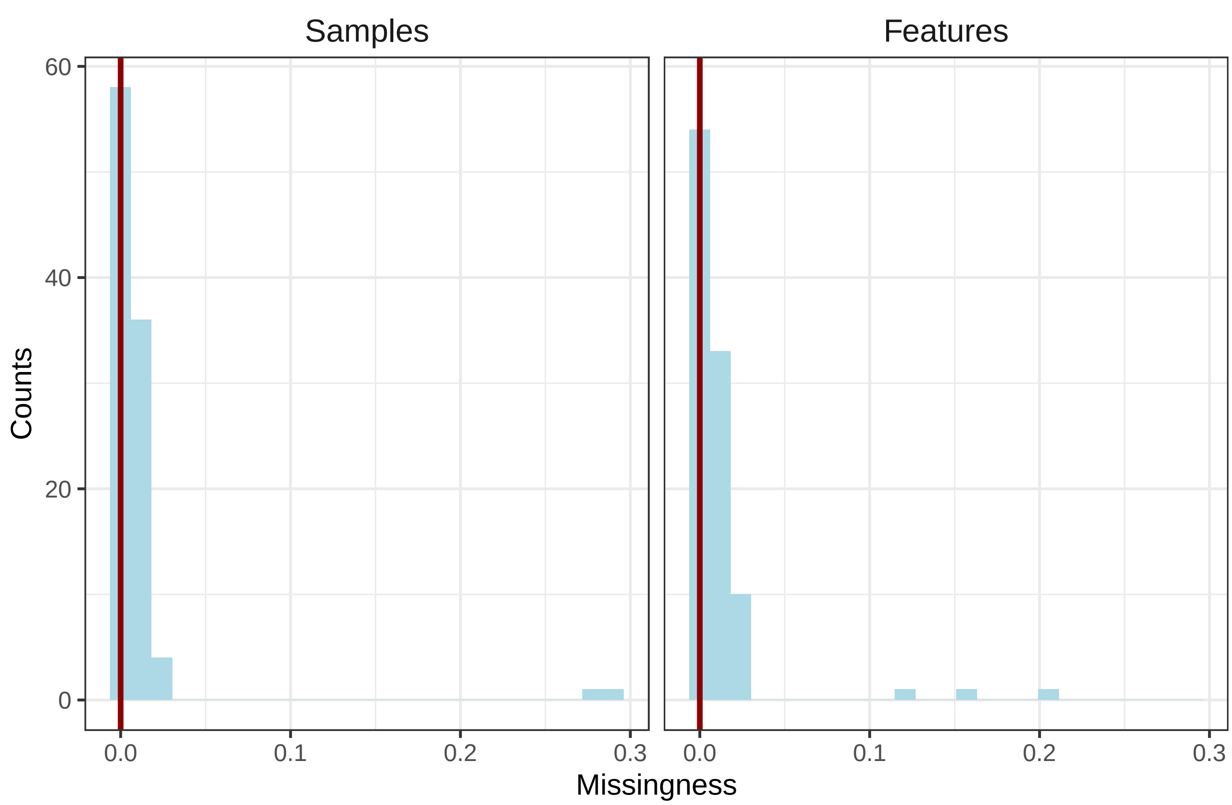
Table 2: Table of sample and feature missingness percentiles.

Percentile	Samples	Features
0.00	0.00	0.00
0.25	0.00	0.00
0.50	0.00	0.00
0.75	0.01	0.01
1.00	0.29	0.20

Table 3: Estimates of samples sizes under various levels of feature missingness.

Missingness (%)	Sample Size
10%	90
20%	80
30%	70
40%	60
50%	50

3.2 Summary of sample and feature missingness



Distribution of missingness with median illustrated by the red vertical line.

Table 4: Sample exclusion summary

Exclusion Reason	**Count**
derived feature	0
xenobiotic feature	98
extreme sample missingness	0
extreme feature missingness	0
user defined sample missingness	2
user defined feature missingness	0
user defined sample totalpeakarea	0
user defined sample pca outlier	0
outlier udist turned na	0
outlier udist winsorized	0

Note:

Six primary data filtering exclusion steps were made during the preparation of the data. User-defined thresholds indicated with an asterisk.

[1] Samples with missingness >=80%.

[2] Features with missingness >=80% (xenobiotics are not included in this step).

[3] Sample exclusions based on the user defined threshold of >=20%.

[4] Feature exclusions based on user defined threshold of >=20%, (xenobiotics are not included in this step).

[5] Samples with a total-peak-area or total-sum-abundance that is >=5 standard deviations from the mean.

[6] Samples that are >= 5 standard deviations from the mean on principal component axis 1 and 2

4 2. Data Filtering

4.1 Exclusion summary

4.2 Next

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